Shyam Sivan
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EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Aerospace Engineering, *Minor in Astrophysics | GPA: 3.43* *May 2028*
Relevant Coursework: Thermodynamics, Fluid Mechanics, Aerodynamics, Mechanics of Materials, Experimental Methods, Differential Equations

EXPERIENCE

Yellow Jacket Space Program Atlanta, GA
Felicette II Vehicle Structures Member – Filament Winder Development September 2026 - Present

- Assisted in designing vehicular structural components for a liquid rocket team aiming to reach the Kármán line, developing filament winder for carbon fiber composite manufacturing to improve consistency and manufacturing efficiency for rocket structures
- Evaluated creel requirements for a 4-axis filament winder designed to process 3 carbon fiber tows simultaneously, addressing tension management concerns during changing payout speeds and carriage reversals through researching variable mass and friction methods
- Compared electronic and mechanical tensioning designs by cost, manufacturing complexity, and spool diameter, supporting design and development of a gravity loaded mechanical creel with friction pads
- Simulated fully mechanical, gravity loaded creel tensioning system under multiple fiber payout conditions, confirming stable system tension with minimal slack and tension spikes in simulated scenarios

Propulsive Landers at Georgia Tech Atlanta, GA
VTVL Hopper Propulsion Member – Injector Development January 2026 - Present

- Collaborated on developing propulsion system for a team aiming to become the first organization to make an untethered VTVL hybrid rocket targeting a 50m hop, supporting development of 1000N engine and flight hardware operating at pressures above 300 bar
- Created propulsion P&IDs to certify physical plumbing and fabrication aligned with ideal system connections before assembling tubing and verifying valve configurations to ensure hardware integration and test readiness
- Developed CAD models for ground test and finalized lander body dimension and configurations, applying GD&T and coordinating interfaces across 4 subteams to define clearance and mounting locations before fabrication
- Analyzed post hot fire data from 3 tests, identifying anomalies with the combustion chamber and notifying the combustion chamber and nozzle teams on hardware adjustments allowing efficient residue purging needed for proper combustion flow
- Compared 4 injector geometries through literature, SolidWorks, and ANSYS Fluent, evaluating atomization behavior, pressure loss, manufacturability, and combustion characteristic differences, selecting a coaxial injector based on manufacturability and flow constraints

RESEARCH

Daedalus Lab at Georgia Tech Atlanta, GA
Injection Molding and Mechanical Recycling Researcher August 2026 – Present

- Researched repeated mechanical recycling cycles' effects on recycled PET and LDPE long-term tensile strength and stress-strain data
- Operated injection molding equipment to produce dog bone specimens for mechanical testing of recycled PET/LDPE material reliability
- Prepared recycled PET and LDPE specimens for characterization through density, viscosity, melt-flow rate, stiffness, and toughness measurements to evaluate how recycling affects mechanical performance, assessing material suitability for repeated reuse

PET Voxel Development Researcher May 2026 – August 2026

- Investigated the mechanical properties of recycled PET-based 3D printing filament through voxel-based specimen design to provide a sustainable and lightweight alternative to current filaments for off-planet structural applications
- Reduced material waste by approximately 20% by optimizing our material extraction device and 3D printer settings through iterative testing across multiple plastic feedstocks and material use-cases, increasing the proportion of feedstock converted to viable material
- Designed CAD fixtures to standardize extraction from recycled plastic pieces and improve structural stability of our extraction device, increasing sample consistency and enabling repeatable structural testing across many experimental batches
- Ran compression tests across PLA, PETG, and PET across 3 scales, with load capacities of up to 1.4 kN for PLA and 1.2kN for our PET prototype, verifying comparable material strengths and appropriate alternative usage of recycled PET for PETG

Computational Combustion Lab at Georgia Tech Atlanta, GA
Combustion Engine CFD Researcher January 2026 – February 2026

- Compared temperature and velocity profiles of simulated engines with reference data, quantifying agreement with published literature to assess model predictions under conditions such as different operating pressures, inlet velocities, and inlet temperatures
- Developed Python scripts to extract temperature and axial velocities from 10 simulation outputs and generate consistent comparison plots, reducing manual post-processing time from 20 to 7 minutes
- Analyzed 3 simulation cases in ParaView to characterize how inlet velocity affected axial velocity and temperature distribution

SKILLS

Software: CAD, GMAT, CFD, FEA, ANSYS Fluent/Mechanical, OpenRocket, MATLAB, Python, Unix/Linux, GitHub, ParaView, LabVIEW, MS Office
Design: GD&T, DFM, Additive Manufacturing, Technical Drawings, Mechanical Testing, Mechanical Assembly, Machine Shop, Trade Studies
Technical: Hybrid Rocket Systems, Data Analysis, Injector Design, Propulsion Systems, Structural Analysis, Combustion & Fluid Dynamics